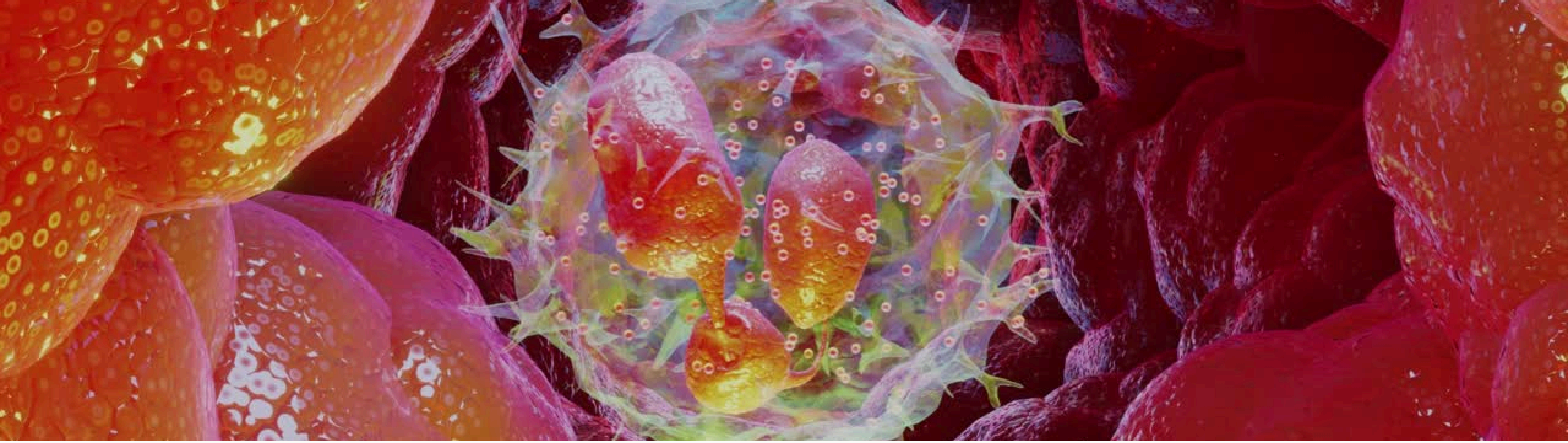


Experiment (3)

WHITE BLOOD CELLS COUNTING AND DIFFERENTIATION

IMMUNOLOGY & SEROLOGY LABORATORY
(BT336)

Introduction

- **White blood cell (WBC)** counting is a crucial component of a complete blood count (CBC) test, which is a routine blood test that provides valuable information about the overall health of an individual.

- The normal range for **white blood cell** count can vary slightly depending on the laboratory and individual factors, but a typical range is approximately **4,500 to 11,000** (cells/µl) white blood cells per microliter of blood.

Low WBC Count

- A low number of **WBCs** is called **leukopenia**. A count of less than 4,500 cells per microliter is below normal.

A lower-than-normal WBC count may be due to:

1. Viral infections
2. Autoimmune disorders
3. Bone marrow problems
4. Cancer-treating drugs, or other medicines
5. Disease of the liver or spleen

High WBC Count

- A higher-than-normal **WBC** count is called **leukocytosis**. A count of more than 11,000 cells per microliter is higher than normal.

A higher-than-normal WBC count may be due to:

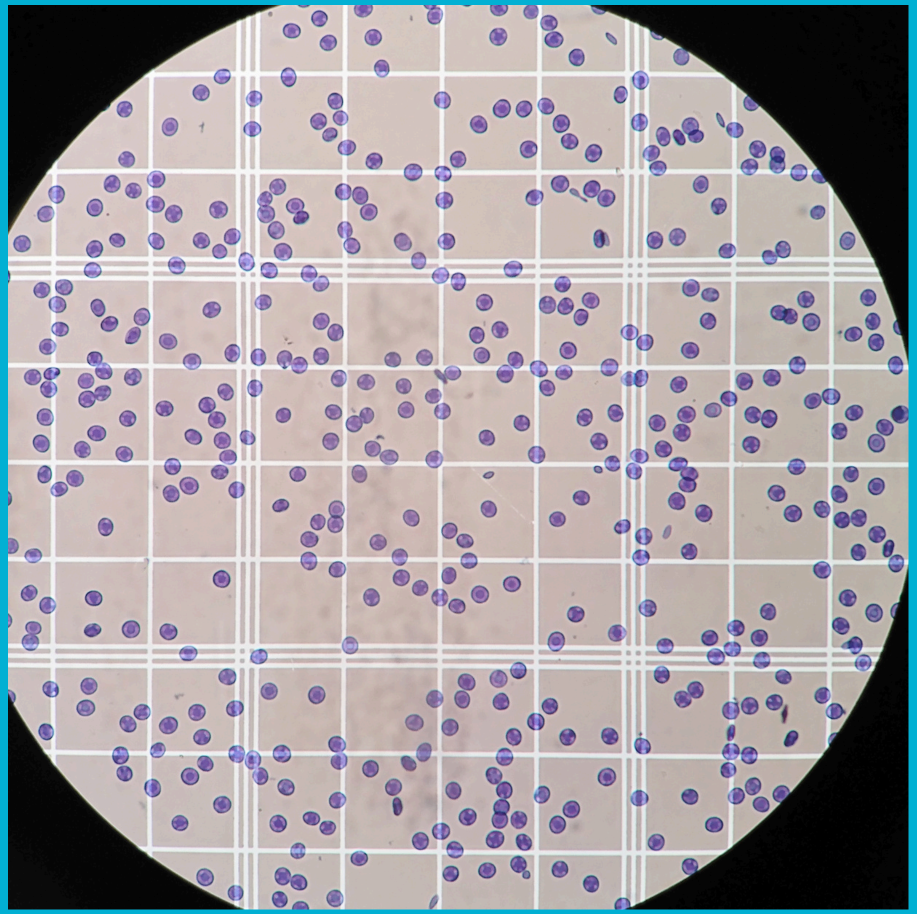
1. Infections
2. Inflammation
3. Certain medical conditions like leukemia and Hodgkin's disease
4. Certain drugs or medicines

Counting Methods

- White blood cell (WBC) counting can be done through various methods, and the choice of method depends on the specific requirements of the laboratory and the information needed.

Here are some common methods for WBC counting:

1. Automated Hematology Analyzers
2. Coulter Counter
3. Flow Cytometry
4. Manual Counting with a Hemocytometer



Automated Hematology Analyzers

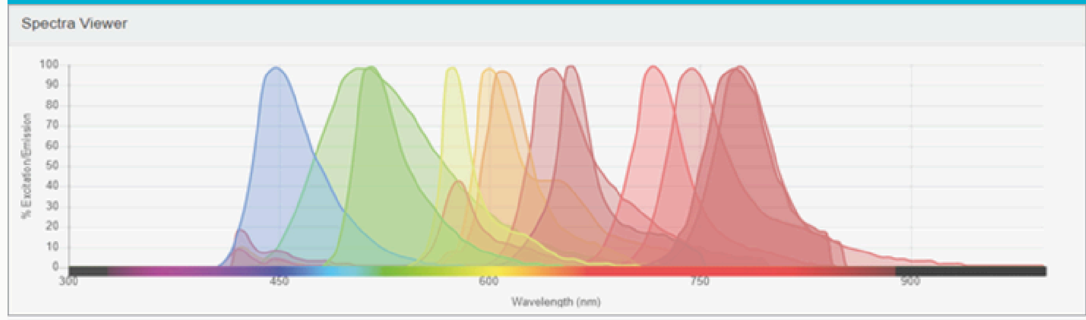
- These are widely used in clinical laboratories for complete blood count (CBC) tests
- Automated analyzers use flow cytometry, electrical impedance, or a combination of both to count and differentiate various blood cells.
- They provide a quick and accurate assessment of **WBC** count and differential (different types of white blood cells).

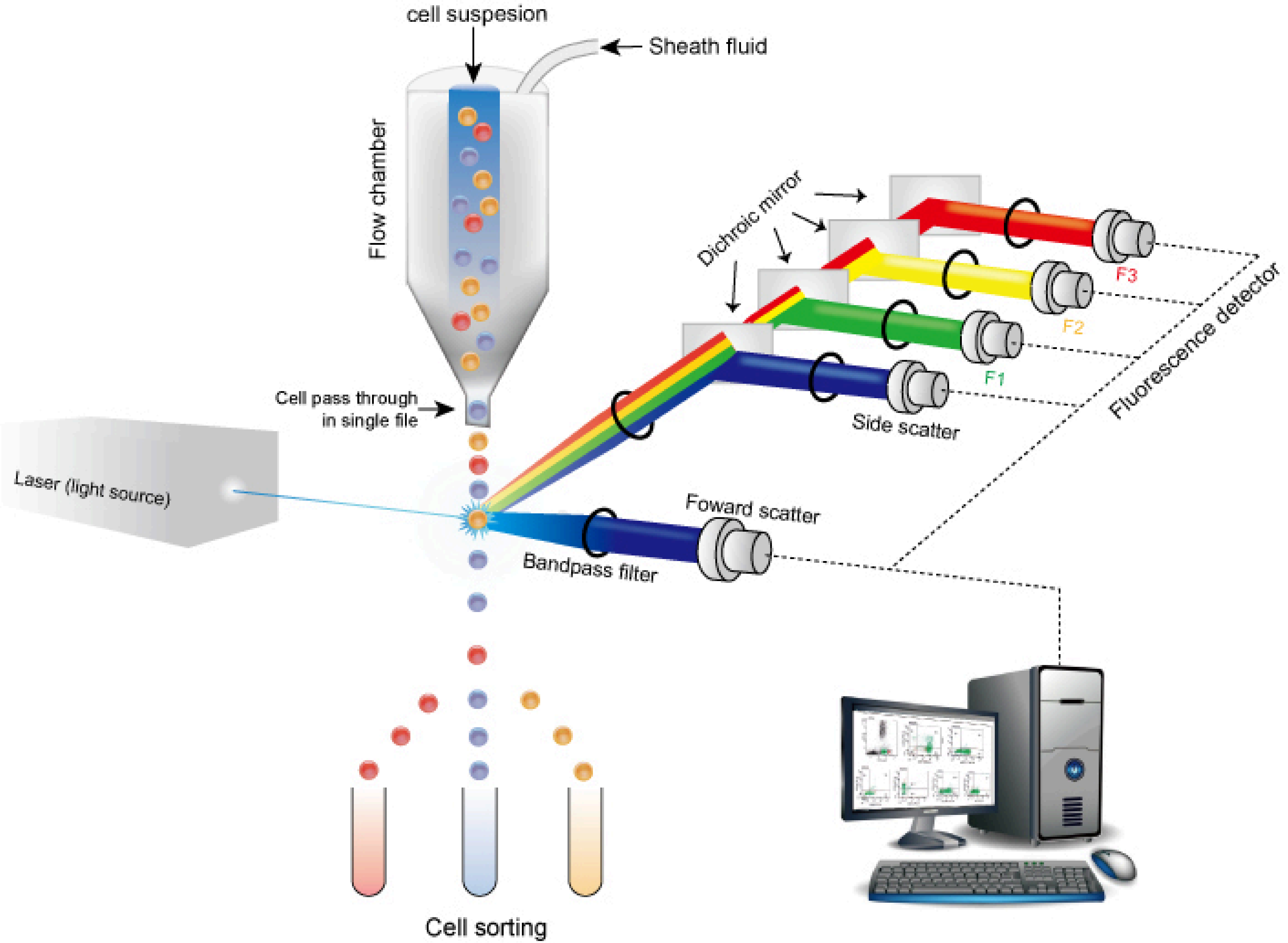
Coulter Counter

- This method relies on electrical impedance to count and size cells as they pass through a small aperture.
- When a cell passes through the aperture, it disrupts an electric current, and the magnitude of the disruption is proportional to the size of the cell.
- Coulter counters are automated and can provide a rapid cell count

Flow Cytometry

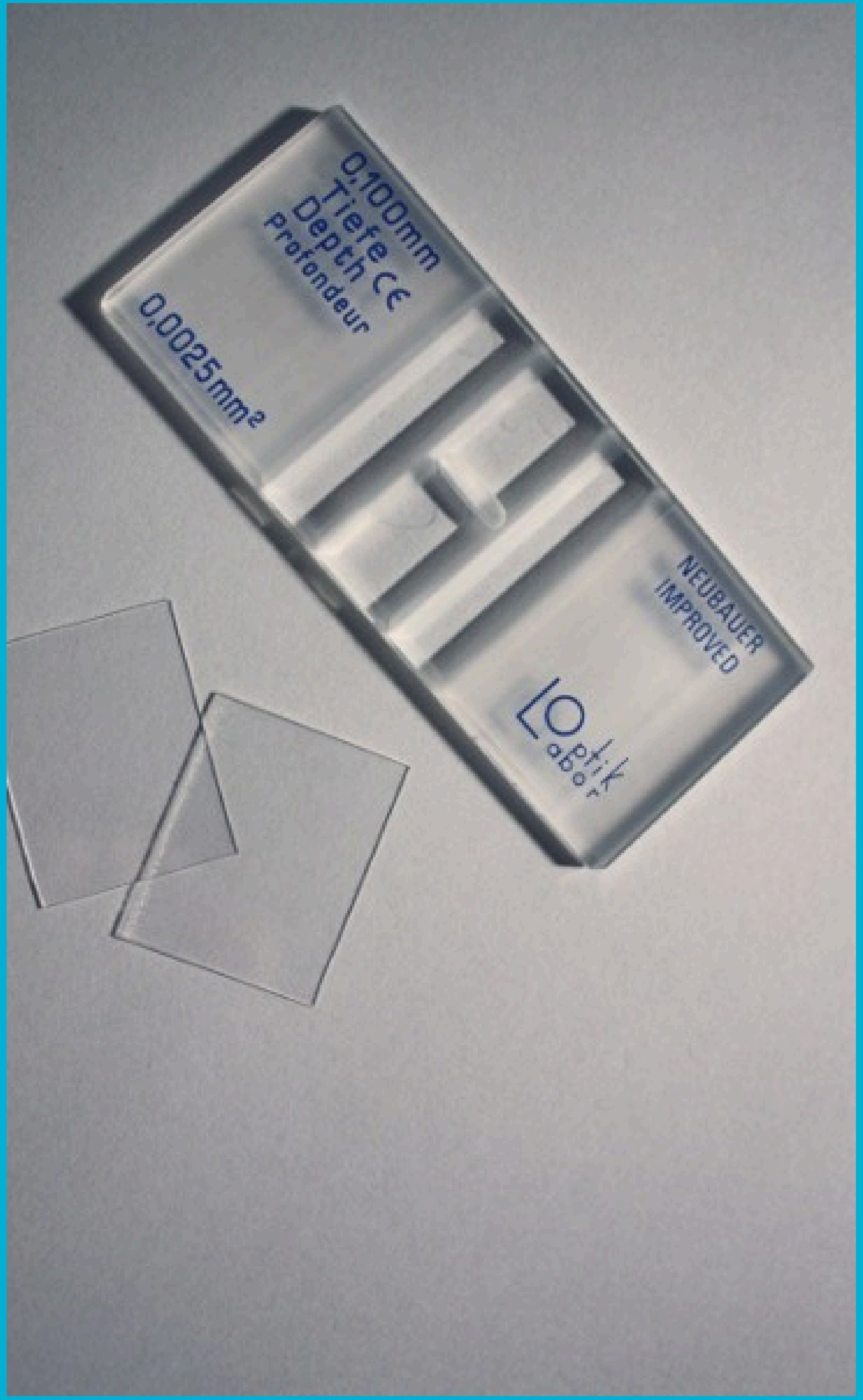
- This method uses laser-based technology to analyze the characteristics of individual cells in a fluid stream.
- It provides detailed information about different cell types based on their size, granularity, and immunological markers.
- Flow cytometry is not only used for counting but also for identifying specific subsets of white blood cells.

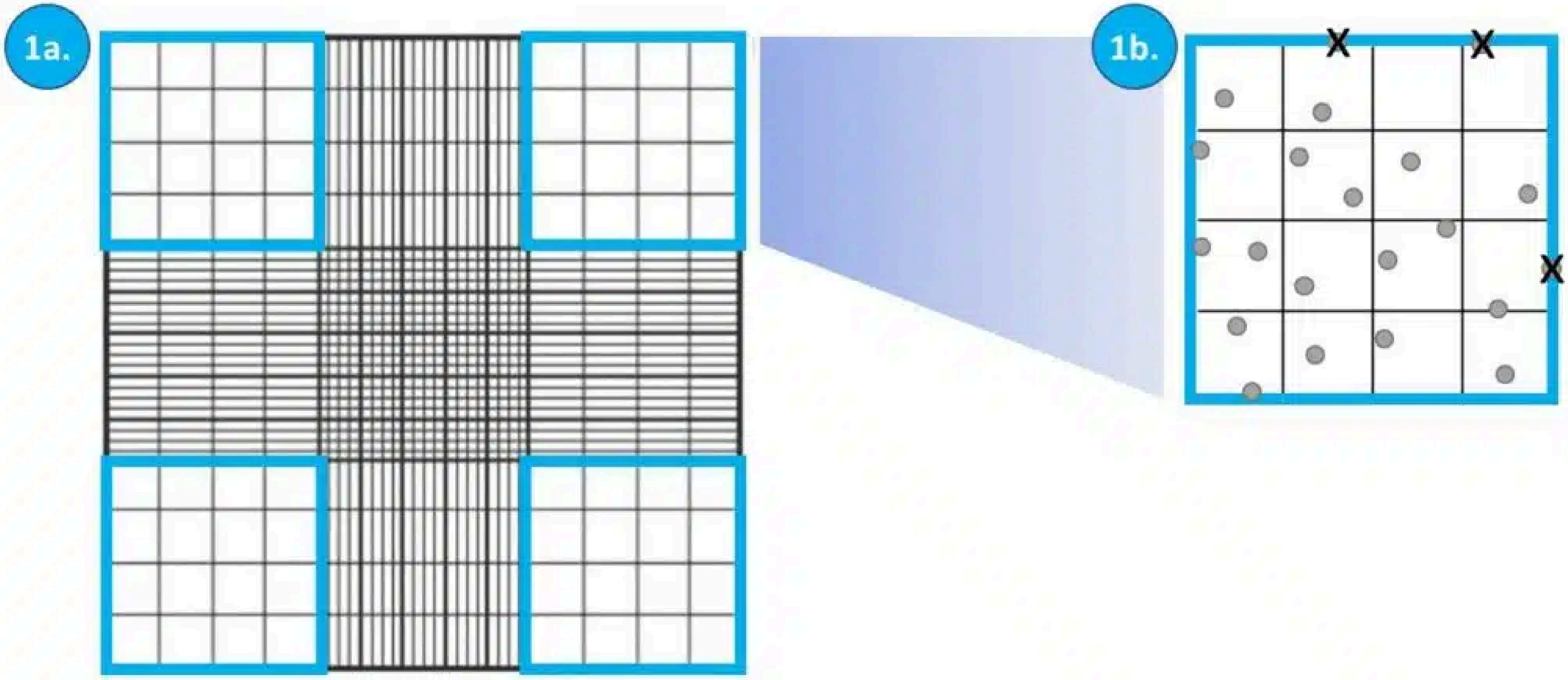



cell suspesion
Sheath fluid
Flow chamber
Dichroic mirror
F3
F2
F1
Fluorescence detector
Side scatter
Cell pass through
in single file
Laser (light source)
Foward scatter
Bandpass filter
Cell sorting

Hemocytometer

- A hemocytometer is a specially designed chamber that allows for manual counting of cells under a microscope.
- Blood is diluted and mixed with a specific stain for better visualization.
- The cells are then counted in a grid on the hemocytometer, and the count is extrapolated to obtain the concentration of WBCs.




1a.
1b.

Hemocytometer Procedure

1. **Prepare Dilution**

- Mix 50 ul of fresh human blood with 950 ul of a 2% acetic acid solution mixed with Trypan Blue.
- Ensure thorough mixing to obtain a homogeneous suspension.

- **The purpose of using 2% Acetic Acid:**

1. Dilute the white blood cells.
2. Lyses all cytoplasmic membranes (the red blood cells and platelets). The diluents also lyse the WBC but take a longer time, so the time factor is critical.

- **The purpose of using Trypan Blue:**
- Trypan blue is a dye used in cell counting, particularly in the assessment of cell viability.
- The dye is taken up by non-viable or dead cells but does not penetrate the membranes of healthy, viable cells.

Hemocytometer Procedure

2. Load Hemocytometer

- Place a clean cover slip on the counting chamber of the hemocytometer.
- Using a pipette or capillary tubes, carefully load the diluted blood into the hemocytometer's counting chamber. Allow the blood to be drawn into the chamber by capillary action

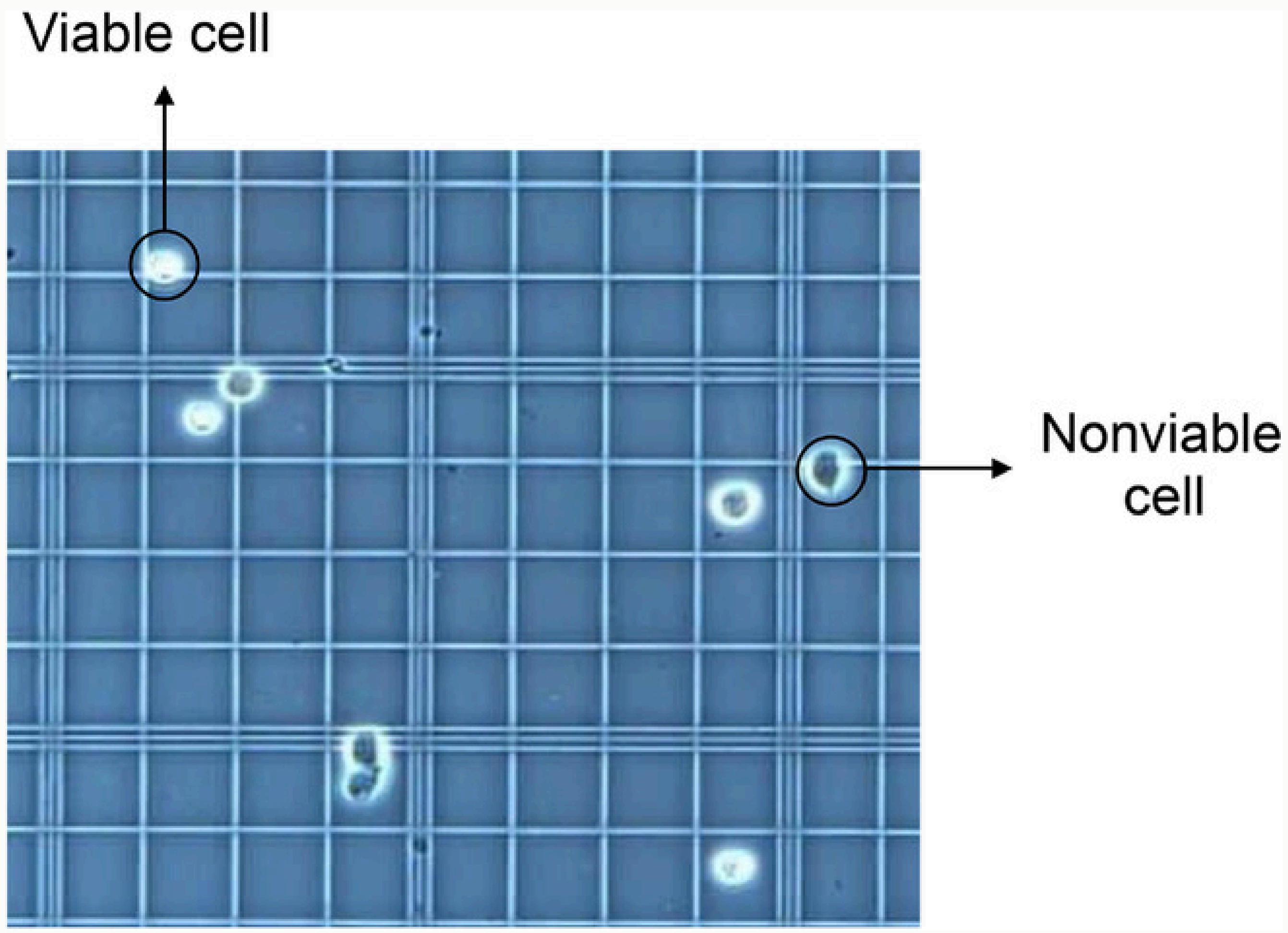
3. Microscopic Examination

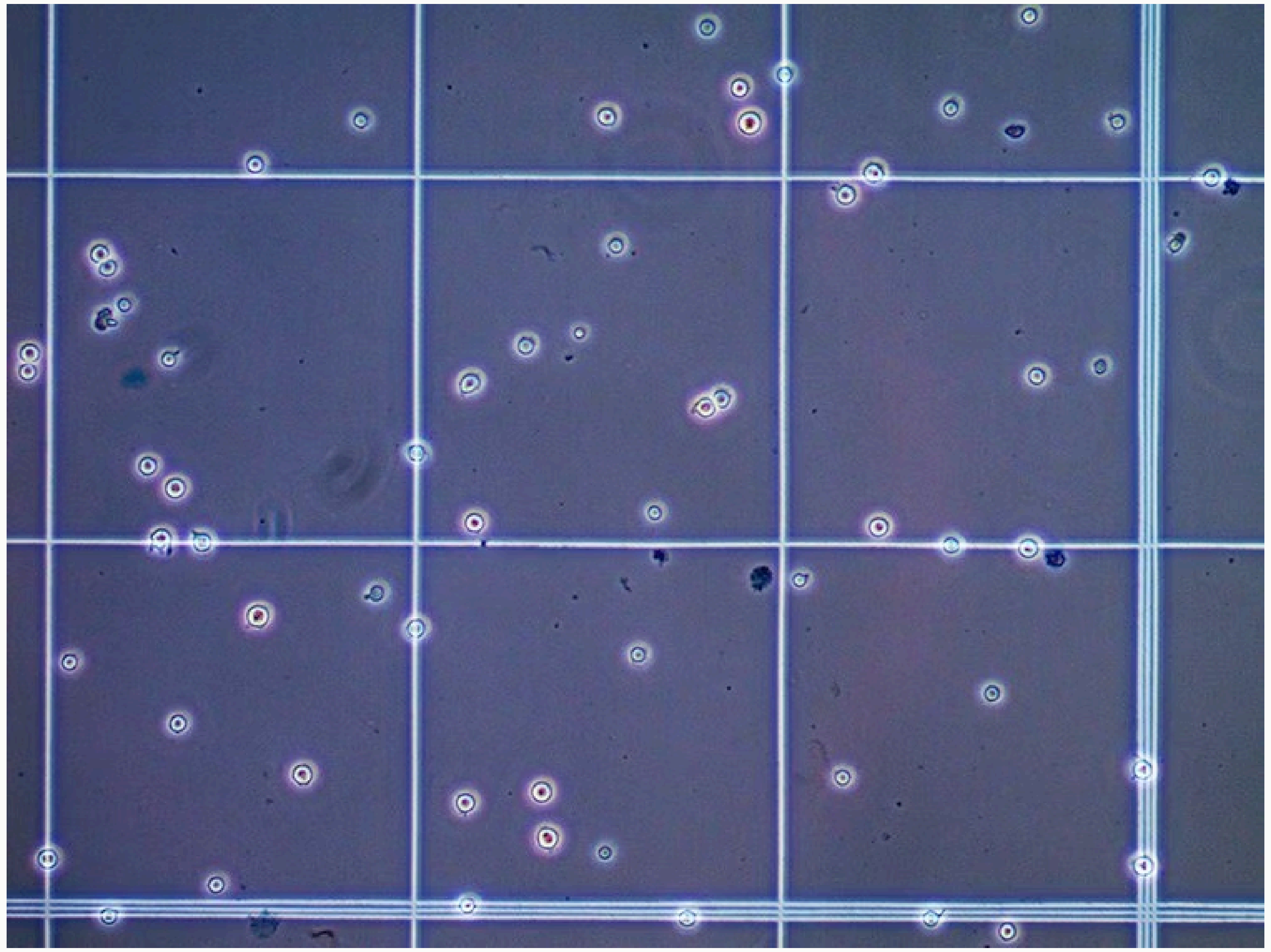
- Place the hemocytometer on the microscope stage and focus on the grid lines.
- Use the 10x or 20x objective lens for counting

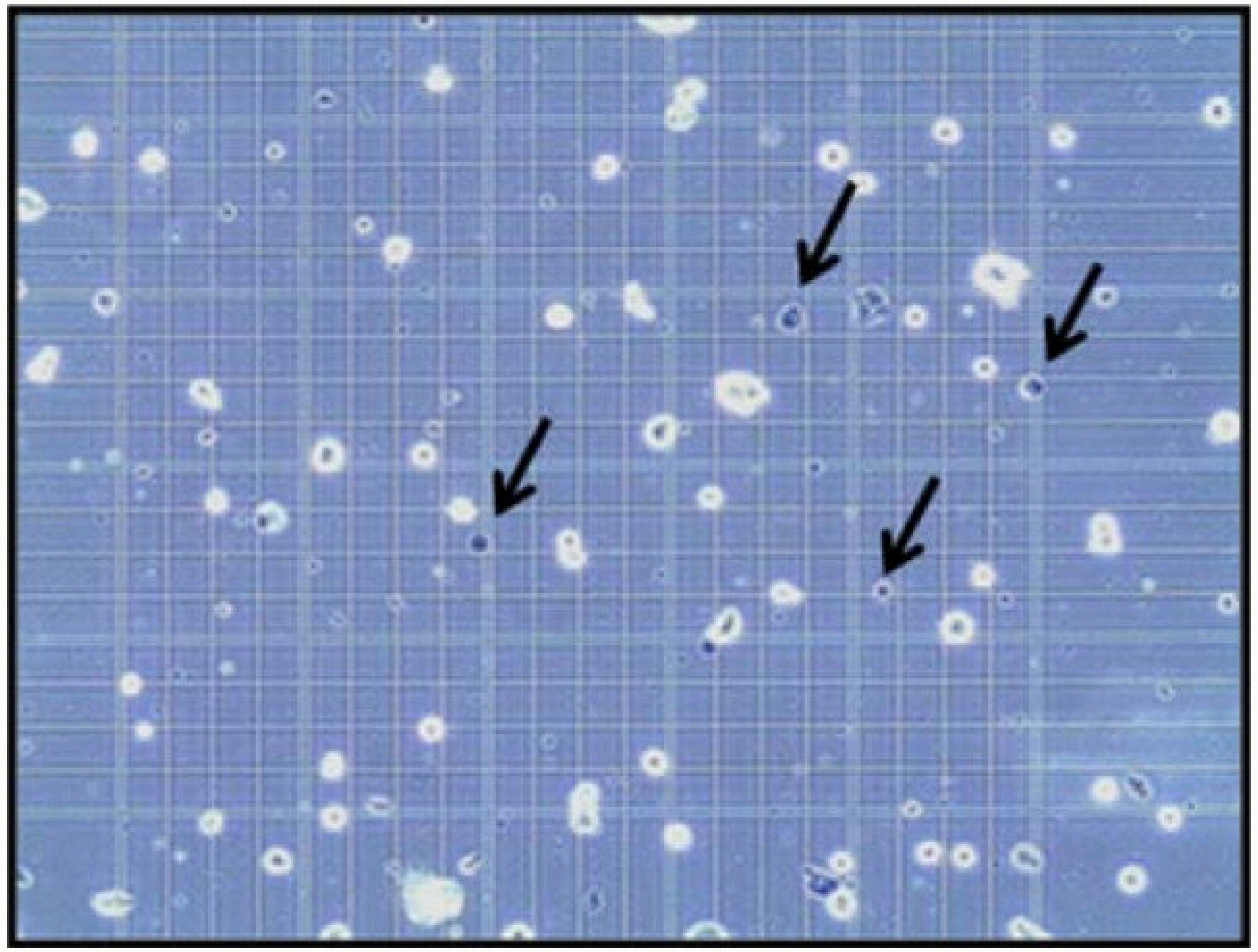
Hemocytometer Procedure

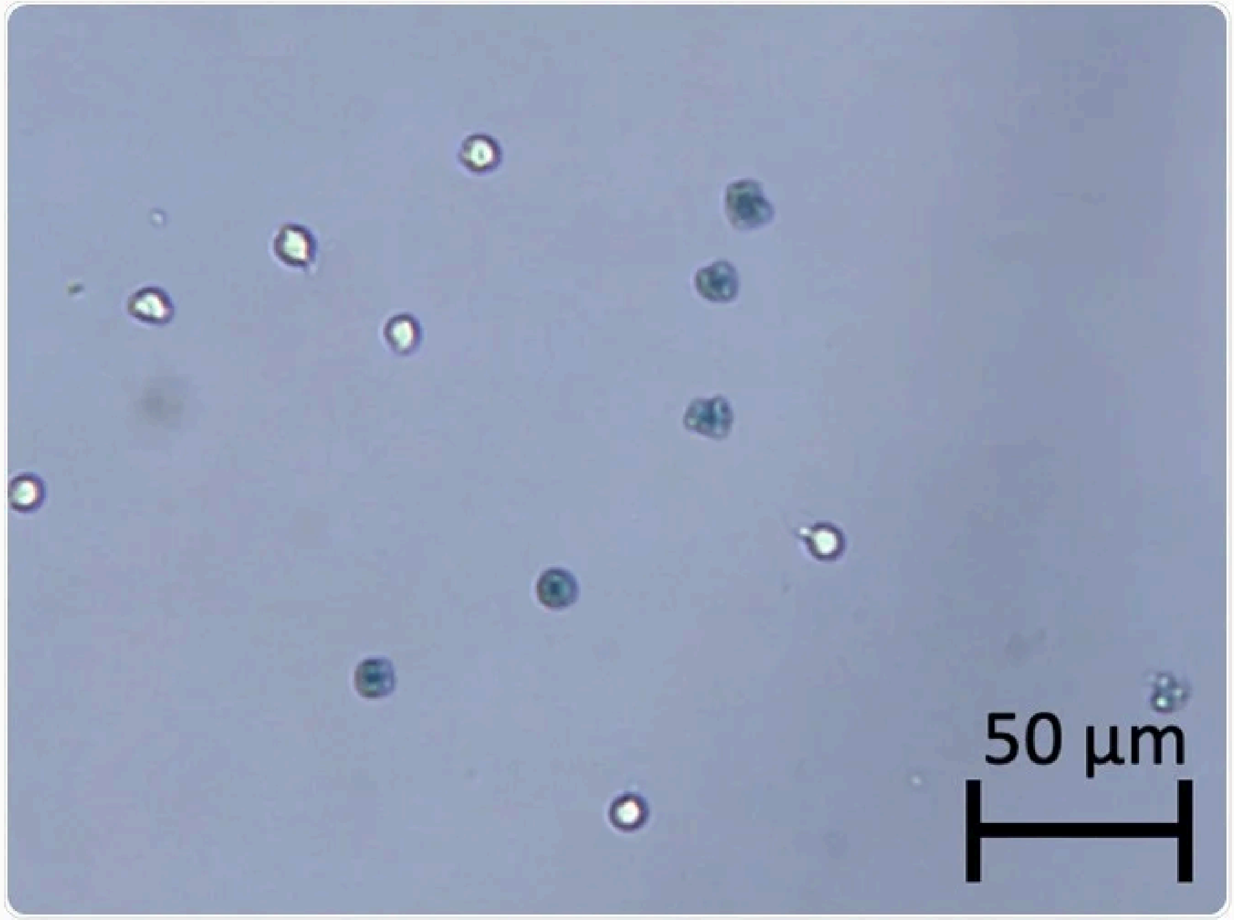
4. Counting Procedure

- Count the number of WBCs in each of the four large corner squares. These squares are marked W1, W2, B1, and B2.
- Each large square is further divided into 16 smaller squares. Count the cells in these smaller squares.
- Count the number of viable (unstained) and non-viable (stained) cells in one or more of the grid squares.
- Exclude cells along the outer borders and corners to avoid duplication.
- Viable cells will appear bright, while non-viable cells will take up the Trypan blue stain and appear blue.

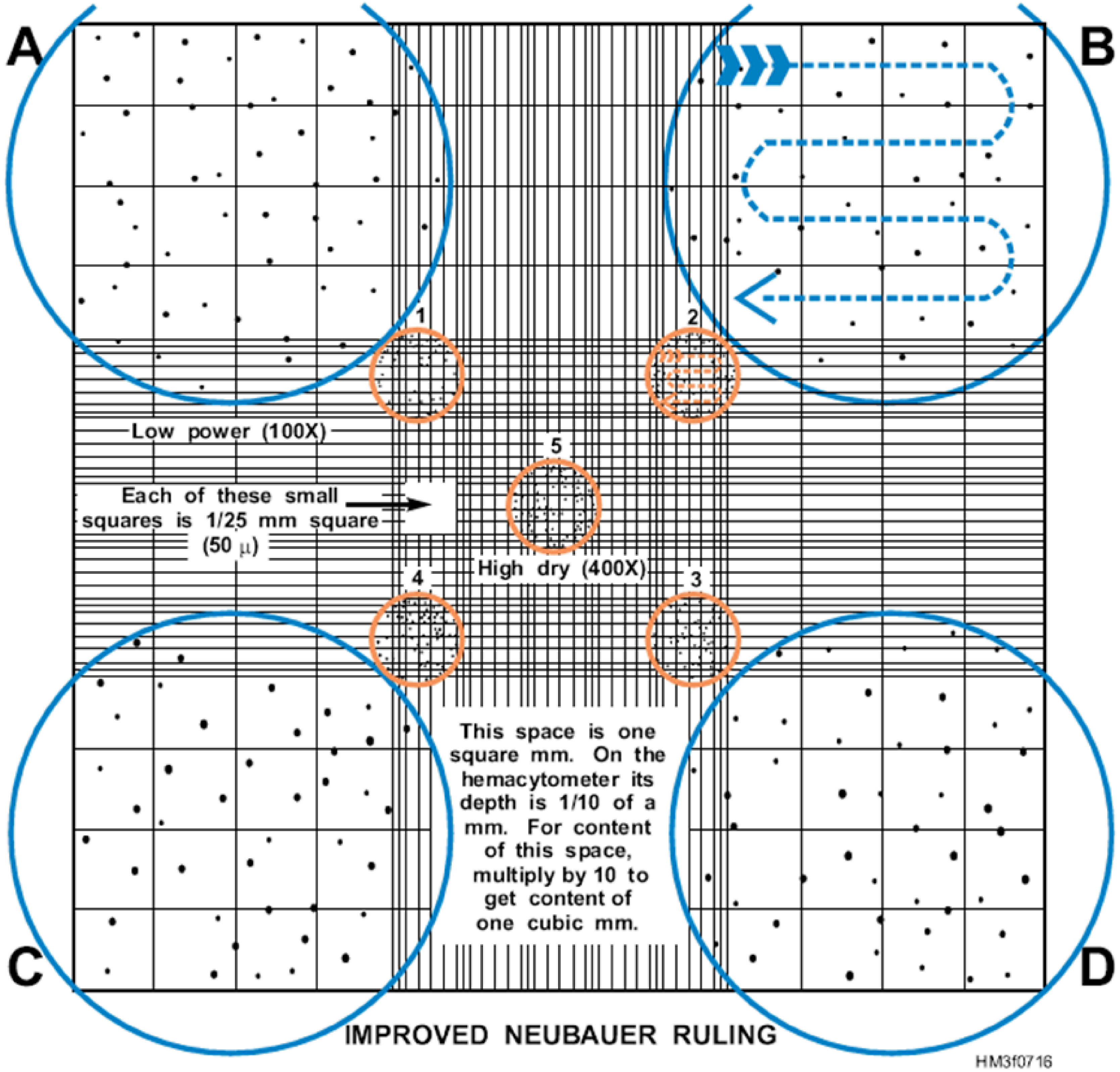

Viable cell
Nonviable cell






50 µm

HEMACYTOMETER (COUNTING CHAMBER)



A - B - C - D ARE FIELDS USED IN DOING THE WHITE BLOOD CELL COUNT.

1 - 2 - 3 - 4 - 5 ARE FIELDS USED IN DOING THE RED BLOOD CELL COUNT.

(Letters, numbers, and arrows are not actually seen in the counting chamber. They are for illustration only. Circles depict areas seen through the microscope.)

Hemocytometer Procedure

5. Calculate WBC Count

- Use the following formula to calculate the total WBC count per microliter:

$$\text{WBC count} = \frac{N}{\text{number of squares counted}} \times \text{dilution factor} \times 10$$

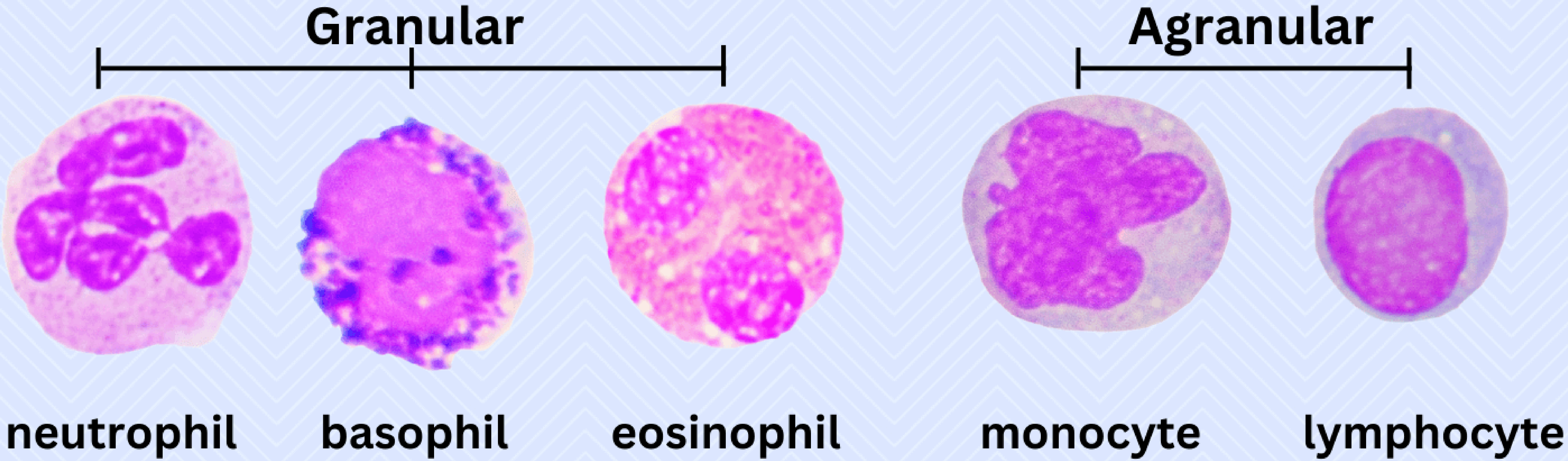
White Blood Cell Differentiation

Leukocytes

(White Blood Cells)

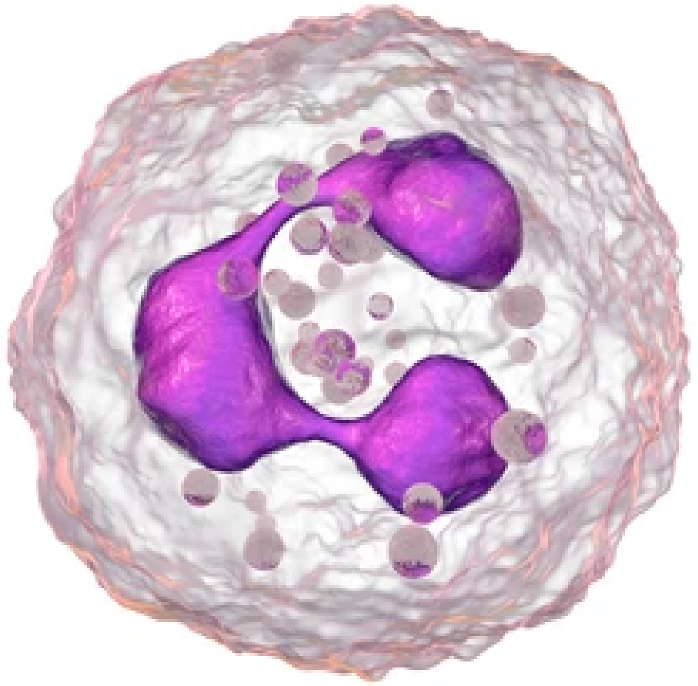
Leukocytes or white blood cells are part of the immune system.

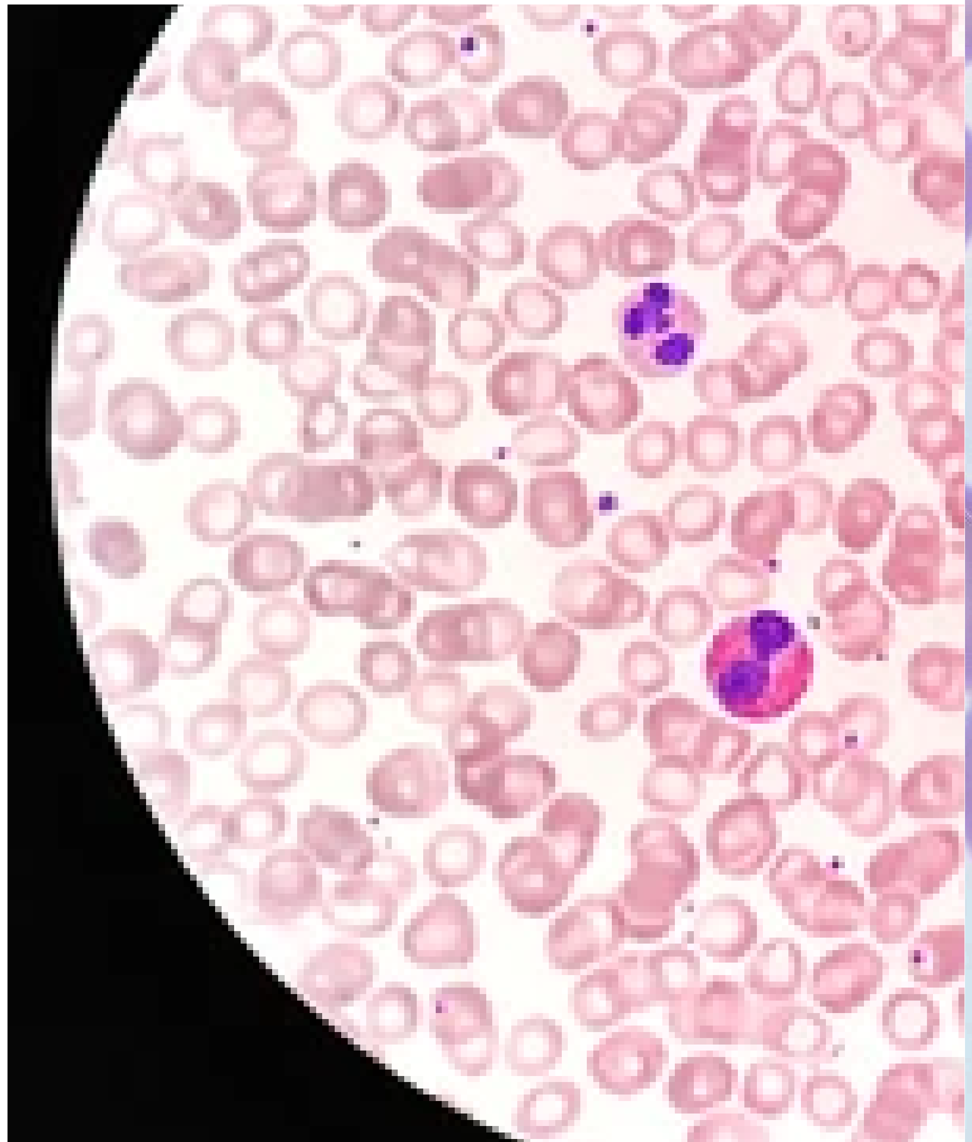
They account for ~1% of blood volume and occur in lymphatic tissue.



NEUTROPHILS

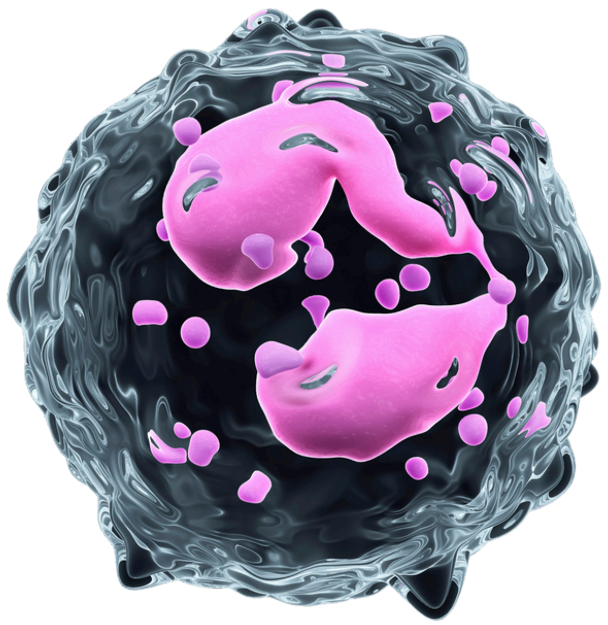
- The nucleus may have two to five lobes connected by thin strands of chromatin.
- Neutrophils contain fine, neutral-staining granules in their cytoplasm.
- 40 - 60% of white blood cells.
- Longevity: a few hours to a few days.
- Function: Neutrophils are the first responders to sites of infection or inflammation and are highly phagocytic. The granules contain enzymes and antimicrobial proteins involved in the destruction of engulfed microorganisms.

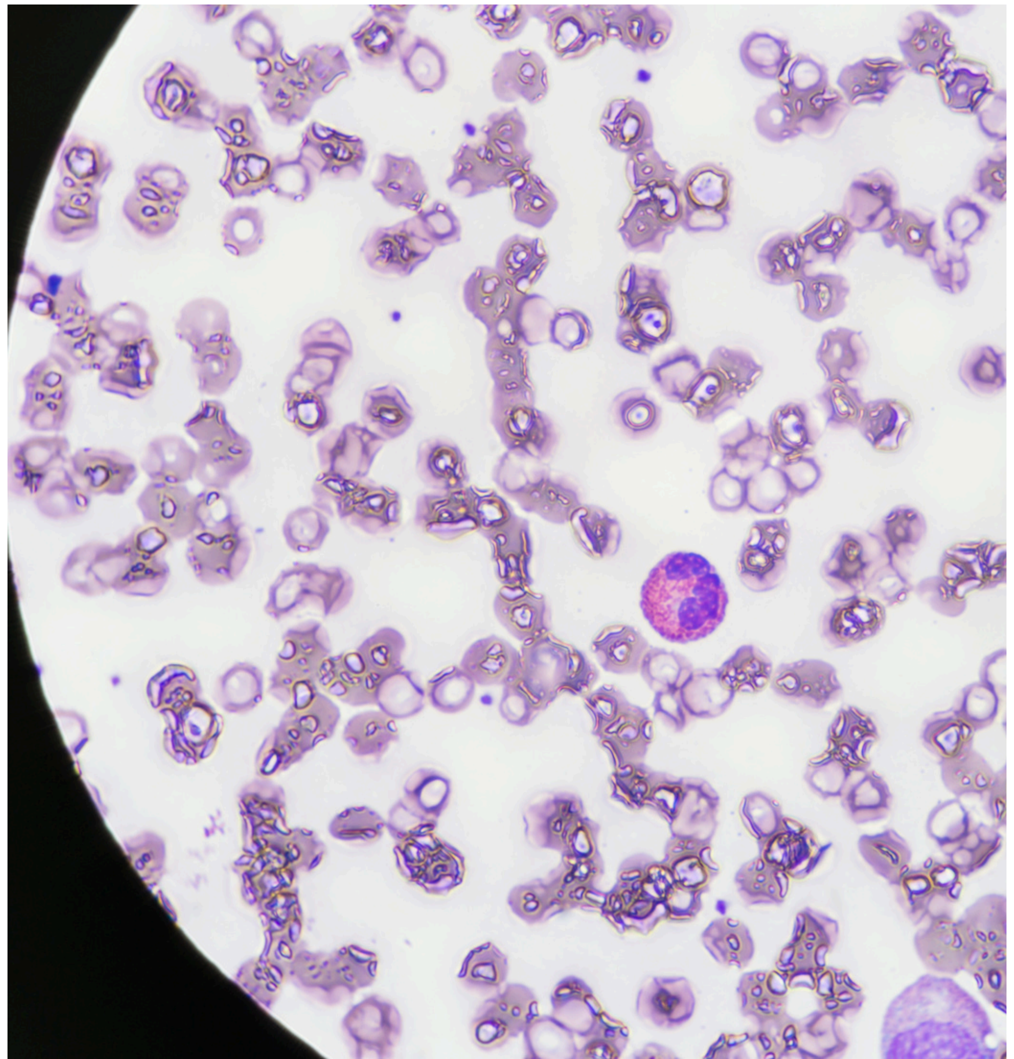


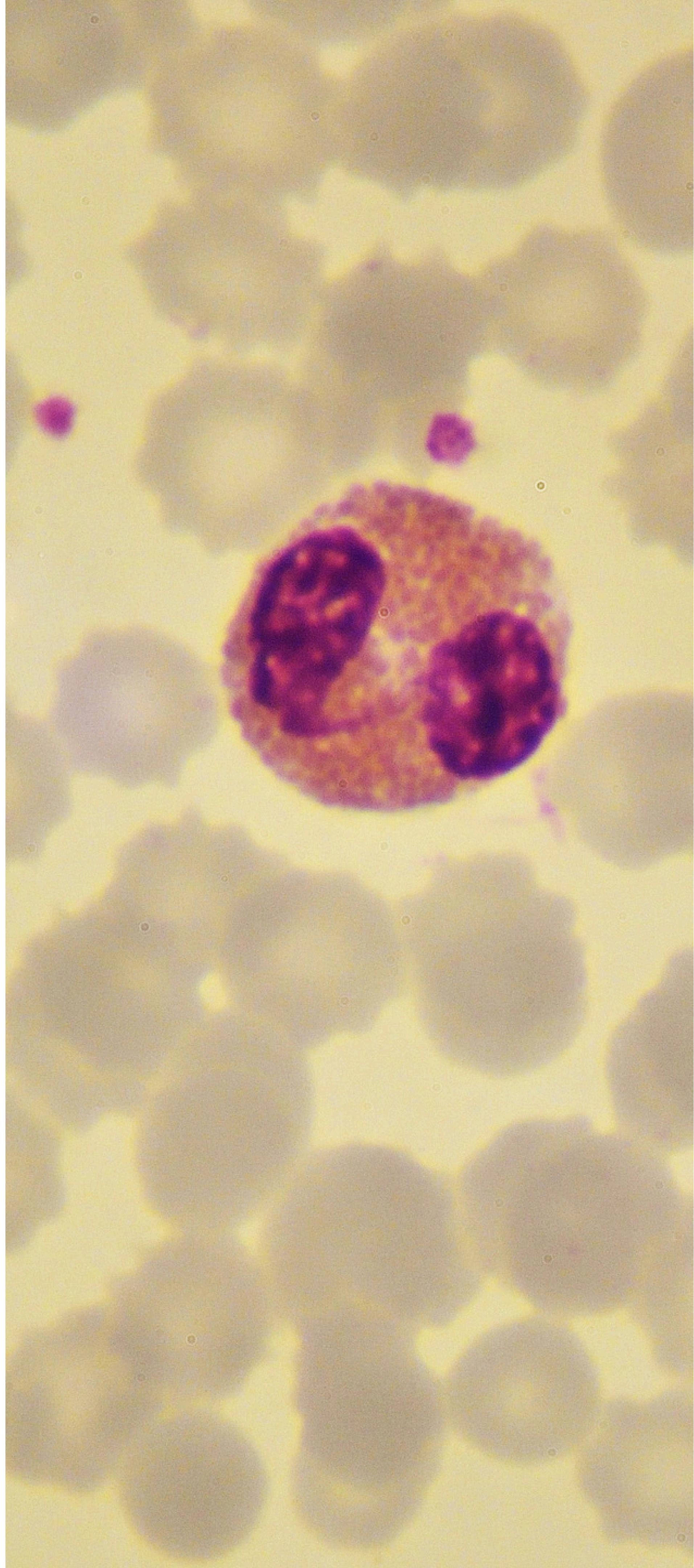


EOSINOPHILS

- They typically have a two-lobed nucleus.
- The granules take up an eosinophilic stain (Acidic), resulting in a red or orange coloration.
- 1 - 4% of white blood cells.
- Longevity: several days to weeks
- Function: Eosinophils are associated with immune responses, particularly those involved in allergic reactions and defense against parasitic infections. The granules contain various enzymes and proteins, including major basic protein (MBP) and eosinophil peroxidase, which play a role in the immune response.







BASOPHILS

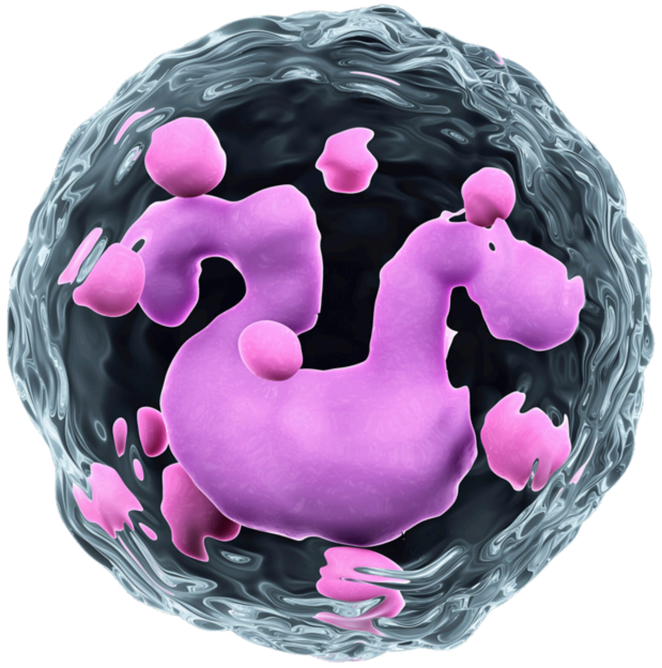
- They have a two-lobed nucleus.

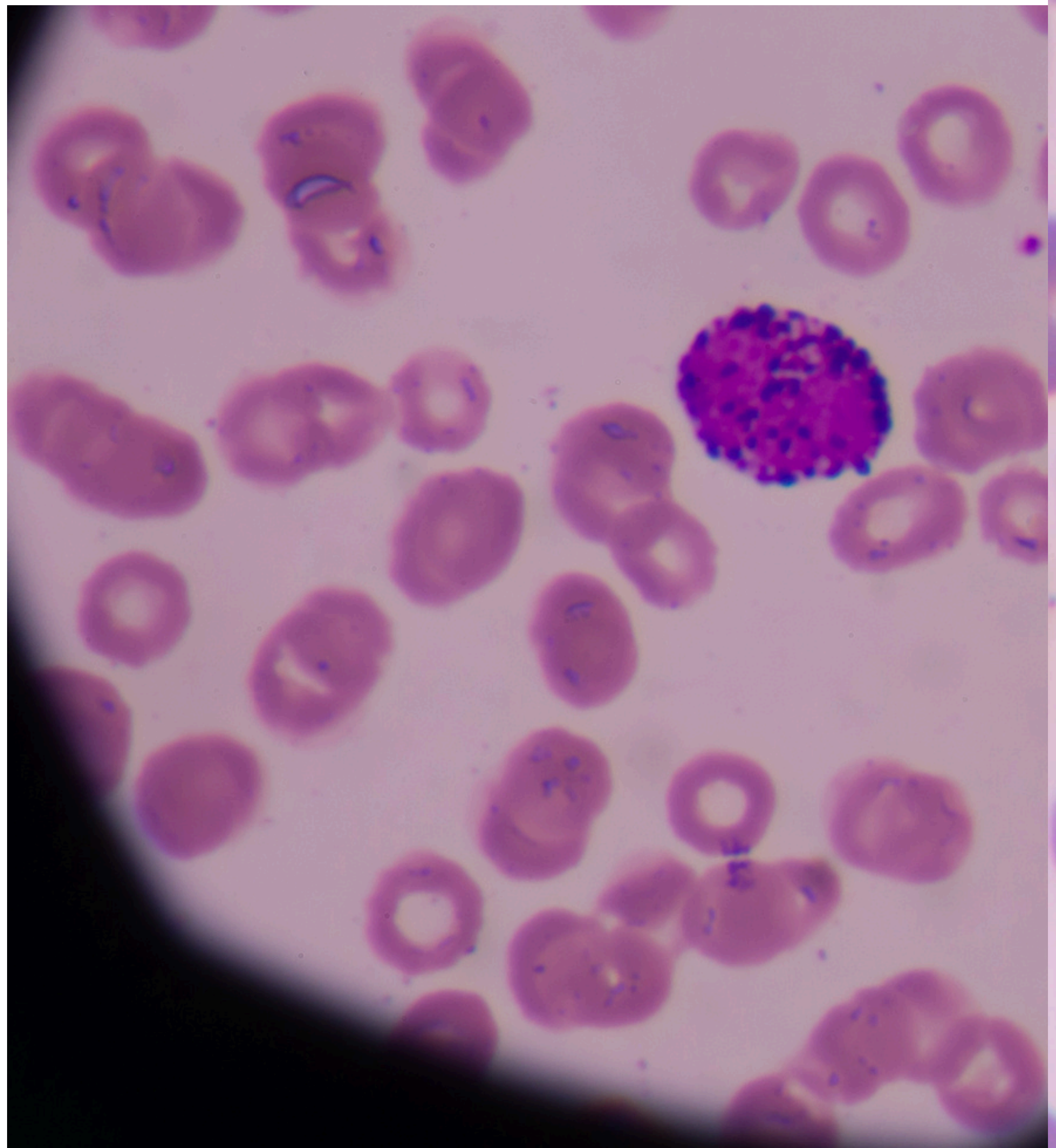
- Basophils have large, coarse, dark-staining granules in their cytoplasm. Basophils often take up a basic stain, such as methylene blue or toluidine blue. The granules stain intensely, giving the cells a deep blue or purple color.

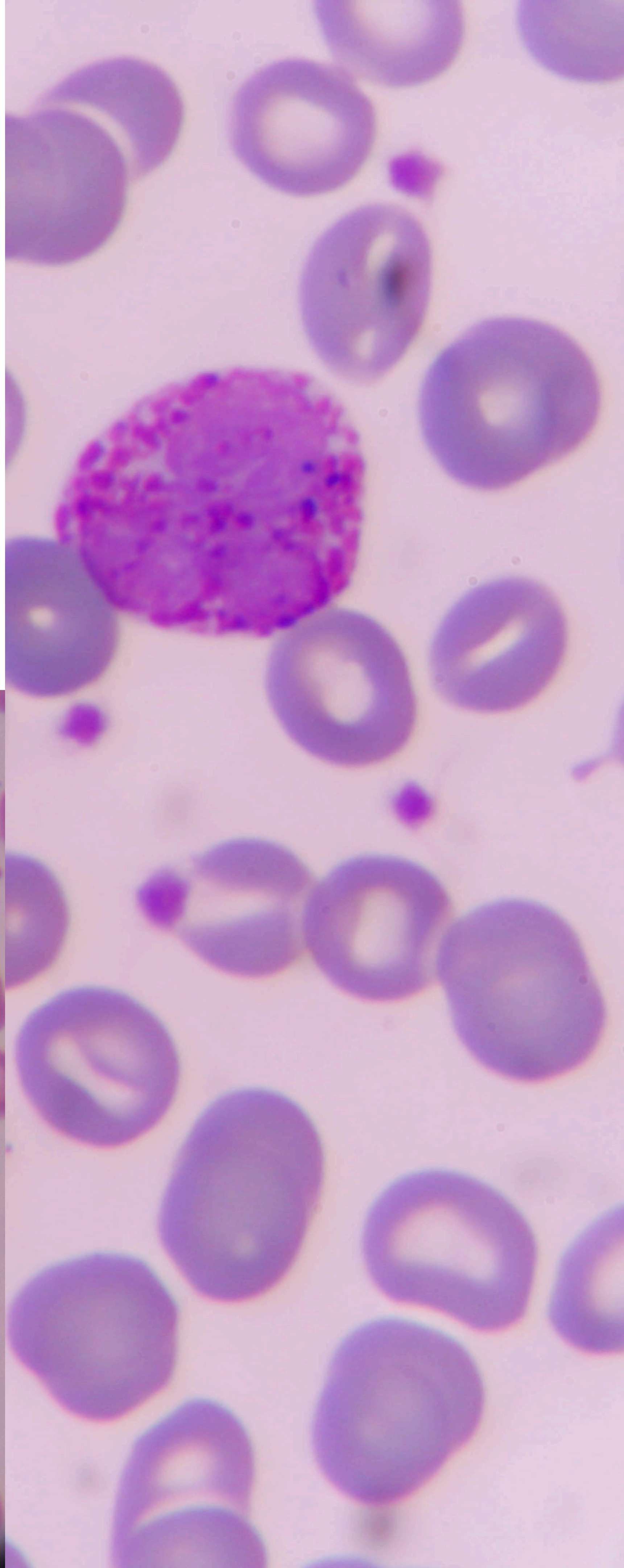
- 0.5 - 1% of white blood cells.

- Longevity: typically measured in days

- Release histamine, heparin, and other inflammatory mediators in response to allergic reactions. They play a role in the body's defense against parasites.







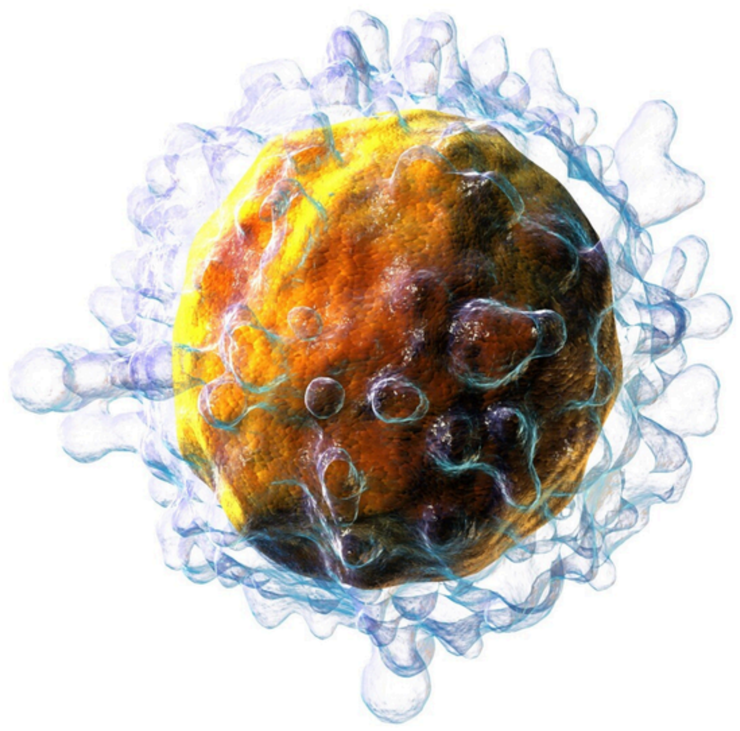
LYMPHOCYTES

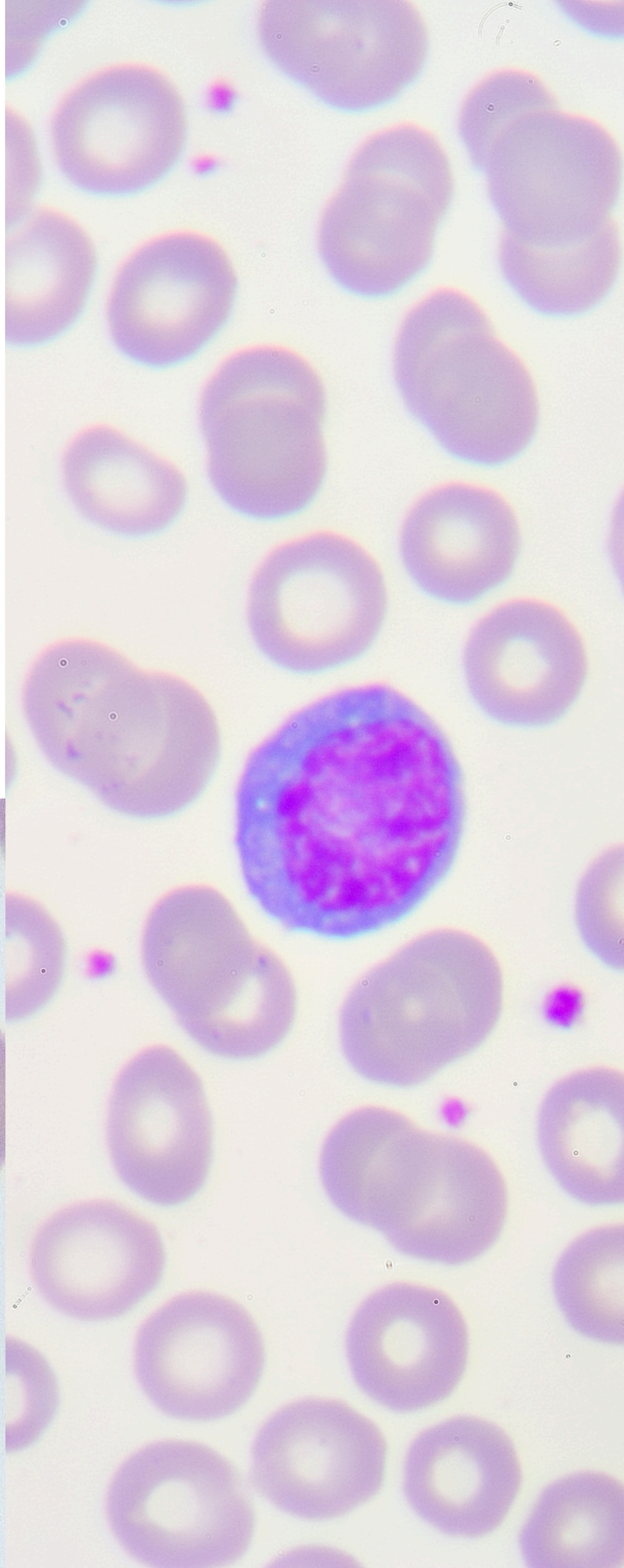
- They have a round or oval-shaped nucleus, which is large in proportion to the cell's size. The nucleus often takes up most of the cell's volume, and the cytoplasm forms a thin, agranular rim around it.

- They stain with both acidic and basic dyes, resulting in a color that varies from light blue to deep purple.

- 20 - 40% of white blood cells.

- Function: Lymphocytes are key components of the adaptive immune system, involved in the recognition and elimination of pathogens.
- There are two main types of lymphocytes: T lymphocytes (T cells) and B lymphocytes (B cells), each with specific roles in immune responses.



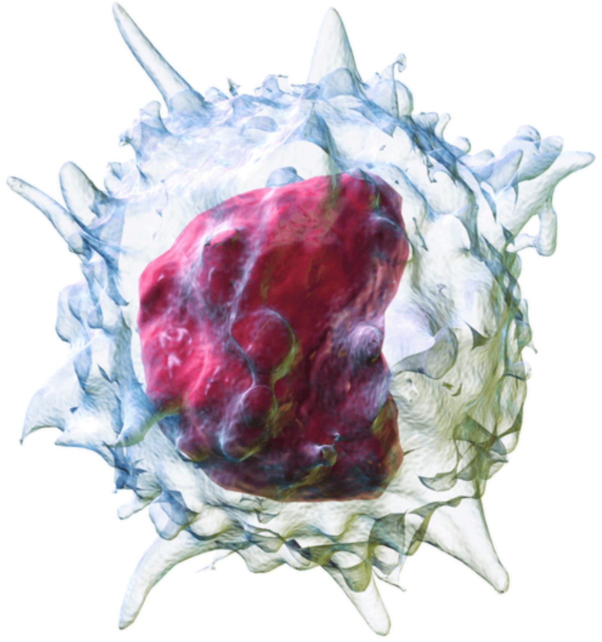


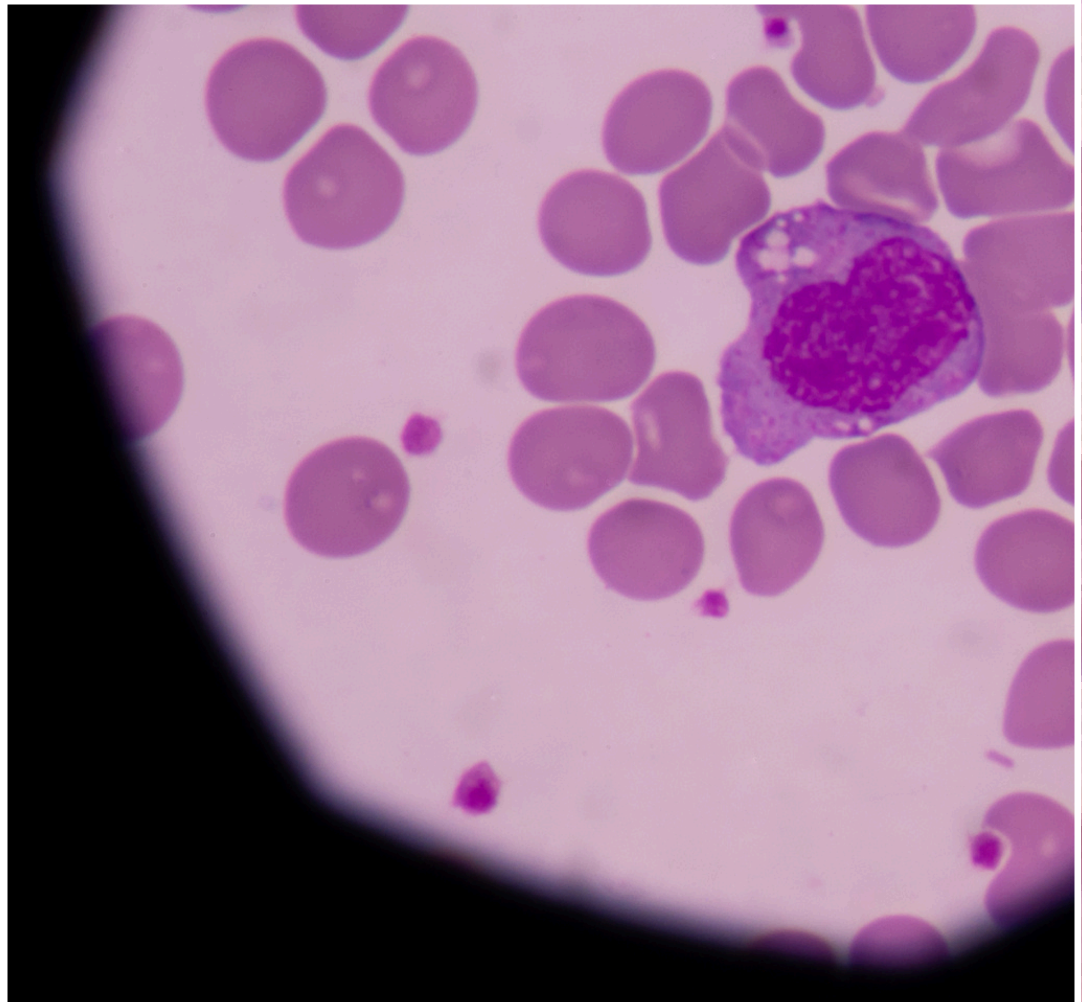
MONOCYTES

- They have a kidney-shaped or horseshoe-shaped nucleus. Monocytes have abundant, agranular cytoplasm that may appear pale under the microscope.

- 2 - 8% of white blood cells

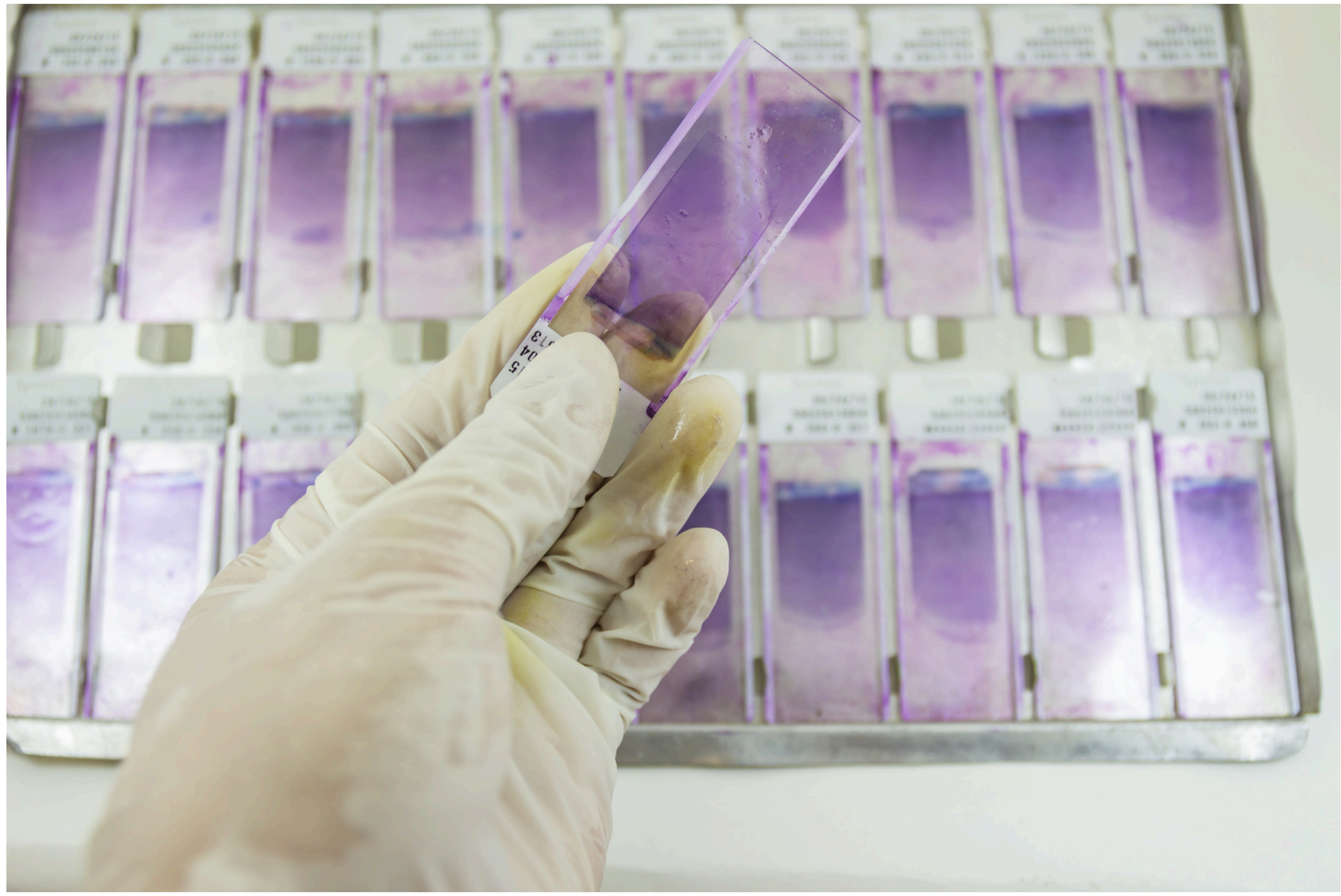
- Function: Monocytes are precursors to macrophages, which are highly phagocytic cells involved in the clearance of cellular debris, pathogens, and foreign substances.





BLOOD FILM PREPARATION

- Blood film preparation, also known as a peripheral blood smear or a blood smear, is a laboratory technique used to create a thin, even layer of blood cells on a glass microscope slide for subsequent microscopic examination.



BLOOD FILM PROCEDURE

1. Spreading the Blood

- Holding the labeled slide at a slight angle, touch the edge of the slide to the drop of blood without letting it touch the skin. The blood will be drawn along the edge of the slide.

2. Creating the Blood Smear:

- Quickly and smoothly spread the drop of blood along the slide's surface using a second clean slide, also known as a spreader slide or feathering slide. This step aims to create a thin, even smear of blood on the slide.
- Adjust the angle between the two slides to control the thickness of the smear. Practice is essential to achieve a proper blood film thickness.

BLOOD FILM PROCEDURE

BLOOD FILM PROCEDURE

3. Drying the Smear

- Allow the blood smear to air dry completely at room temperature. Avoid heat sources or direct sunlight, as they may distort the blood cells.

4. Staining

- Stain the fixed blood smear with an appropriate stain, such as Wright's stain or Giemsa stain, Incubate the smear in the Wright stain for about 5 to 10 minutes. This allows the stain to interact with cellular components, providing contrast for microscopic examination.
- After staining, rinse the smear gently with buffered water or distilled water to remove excess stain.

5. Microscopic Examination

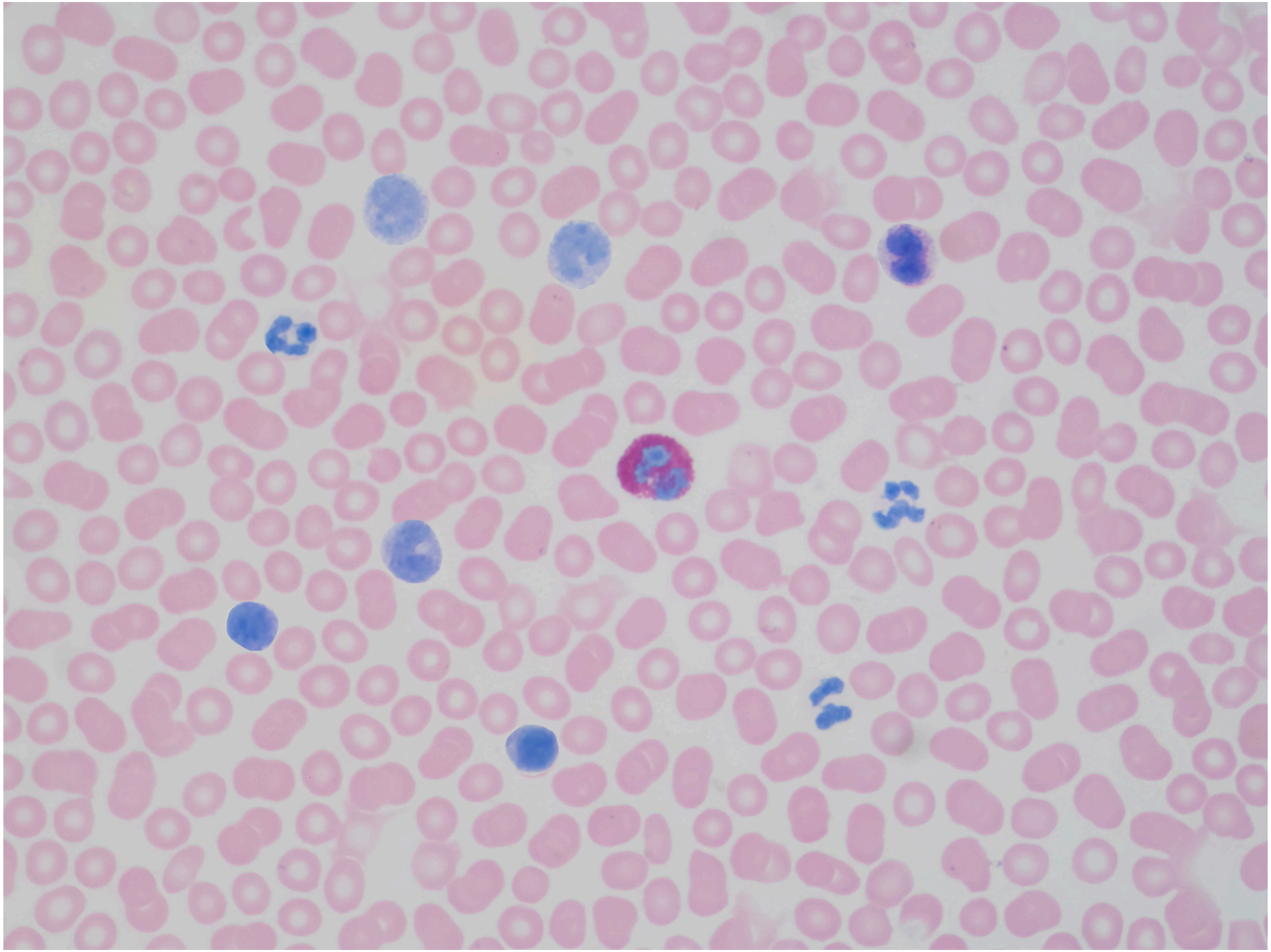


Table 2. Normal White Blood Cell Distribution

White blood cell line	*Normal percentage of total leukocyte count*
Neutrophils	40 to 60
Lymphocytes	20 to 40
Monocytes	2 to 8
Eosinophils	1 to 4
Basophils	0.5 to 1

Information from reference 8.

CANVA SLIDES

https://www.canva.com/design/DAF-CDSgy84/Gchgz-zi5TqIG63TC7WMhw/edit?utm_content=DAF-CDSgy84&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton


THE END!!!